

Test 1 - Funcții complexe

① Rezolvați ecuația cu coef $\in \mathbb{C}$

a) $z^2 - 2(3+2i)z + 2+8i = 0, z = a+bi$

Se obs. că ec. dată este de grad $\Rightarrow$

$$\Delta = b^2 - 4ac = [2(3+2i)]^2 - 4(2+8i)$$

$$= 2(9 + 12i + 4i^2) - 8 - 32i$$

$$= 18 + 24i - 16 - 8 - 32i$$

$$= -12 + 16i$$

$$\sqrt{\Delta} = a+bi \Rightarrow \Delta = (a+bi)^2 \Rightarrow a^2 + 2abi - b^2$$

$$\Rightarrow \begin{cases} a^2 - b^2 = 12 \\ 2ab = 16 \end{cases} \Rightarrow \begin{cases} a^2 - b^2 = 12 \\ ab = 8 \Rightarrow a = \frac{8}{b} \end{cases} \Rightarrow \begin{cases} \frac{64}{b^2} - b^2 = 12 \\ b^2 \end{cases}$$

$$\Rightarrow 64 - b^4 = 12b^2 \Rightarrow b^4 + 12b^2 - 64 = 0, b^2 = t, t \geq 0$$

$$\Rightarrow t^2 + 12t - 64 = 0 \Rightarrow t_{1/2} = \frac{-12 \pm \sqrt{144 + 256}}{2}$$

$$\Rightarrow t_{1/2} = \frac{-12 \pm \sqrt{400}}{2} \Rightarrow t_{1/2} = \frac{-12 \pm 20}{2}$$

$$\Rightarrow t_1 = \frac{-12 + 20}{2} = \frac{8}{2} = 4 > 0 \Rightarrow \boxed{b = \pm 2}$$

$$\Rightarrow t_2 = \frac{-12-20}{2} = \frac{-32}{2} = -16 < 0, \text{ non valide.}$$

$$b=2 \Rightarrow a = \frac{8}{2} = 4, \sqrt{\Delta} = 4+2i$$

$$b=-2 \Rightarrow a = \frac{8}{-2} = -4, \sqrt{\Delta} = -4-2i$$

$$z_{1/2} = \frac{2(3+2i) \pm (4+2i)}{2}$$

$$\Rightarrow z_1 = \frac{2(3+2i) + 4+2i}{2} \Rightarrow z_1 = \frac{6+4i+4+2i}{2}$$

$$= \frac{10+6i}{2} = \boxed{5+3i}$$

$$\Rightarrow z_2 = \frac{2(3+2i) - 4-2i}{2} \Rightarrow z_2 = \frac{6+4i-4-2i}{2}$$

$$= \frac{2+2i}{2} = 1+i$$

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4) Aflați pct. în care urm. fct sunt deriva-
bile și să se calc. deriv. fct. în pct. aflate.

a) $f(z) = x^2 + y^2 + i(3xy - 5x)$

$$\left. \begin{aligned} \frac{\partial P}{\partial x} &= 2x & \frac{\partial Q}{\partial y} &= 3x \\ \frac{\partial Q}{\partial x} &= 3y - 5 & -\frac{\partial P}{\partial y} &= -2y \end{aligned} \right\} \Rightarrow \begin{aligned} 2x &= 3x \\ 3y - 5 &= -2y \end{aligned} \Rightarrow \begin{aligned} x &= 0 \\ y &= 1 \end{aligned}$$

$\Rightarrow f$ deriv în $z_0 = i$

$$\begin{aligned} f'(z_0) &= \frac{\partial P}{\partial x} \Big|_{z_0} + i \frac{\partial Q}{\partial x} \Big|_{z_0} = 2i + \cancel{3i} + i(3i - 5) \\ &= \cancel{2i - 3 - 5i} - 3 - 3i = 2 \cdot 0 - 2i = -2i \end{aligned}$$

$$b) f(z) = 2z^2 - z\bar{z} - \bar{z}^2 - 3z + i\bar{z}$$

$$f(z) = 2(x+iy)^2 - 2(x-iy) - (x-iy)^2 - 3(x+iy) + i(x-iy)$$

$$= 2(x^2 + xiy - y^2) - 2x + 2iy - (x^2 - 2xyi - y^2) - 3x - 3iy + xi + y$$

$$= 2x^2 + 2xyi - 2y^2 - 2x + 2iy - x^2 + 2xyi + y^2 - 3x - 3iy + xi + y$$

$$= x^2 + 4xyi + y^2 - 5x - iy + xi + y$$

$$f(z) = 2(x+iy)^2 - (x+iy)(x-iy) - (x-iy)^2 - 3(x+iy) + i(x-iy)$$

$$= 2(x^2 + 2xyi - y^2) - (x^2 + y^2) - (x^2 - 2xyi - y^2) - 3x - 3iy + ix + y$$

$$= 2x^2 + 4xyi - 2y^2 - x^2 - y^2 - x^2 + 2xyi + y^2 - 3x - 3iy + ix + y$$

$$= -2y^2 - 3x + y + 6xyi - 3iy + ix$$

$$= -2y^2 - 3x + y + i(6xy - 3y + x)$$

$$\frac{\partial P}{\partial x} = -3 \quad \frac{\partial Q}{\partial y} = 6x - 3$$

$$\frac{\partial P}{\partial y} = -4y + 1 \quad \frac{\partial Q}{\partial x} = -6y + 1$$

$$\Rightarrow \begin{cases} -3 = 6x - 3 \\ -4y + 1 = -6y + 1 \end{cases} \Rightarrow \begin{cases} 6x = 0 \\ 2y = -2 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = -1 \end{cases} \Rightarrow z = -i$$

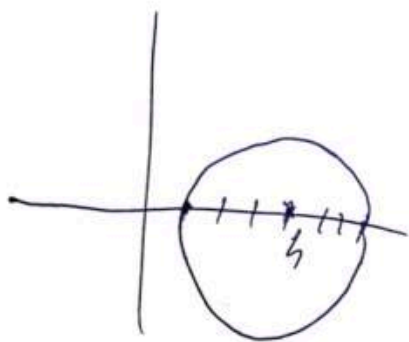
$\Rightarrow f$ deriv in $z_0 = -i$

$$f'(z_0) = \frac{\partial P}{\partial x} \Big|_{z_0} + i \frac{\partial Q}{\partial x} \Big|_{z_0} = -3 + i(-6+1) = -3-5i$$

$$-2 \dots \dots 1: \bar{z} - z + \bar{z}$$

$$c) \int_{|z-4|=3} \frac{2z+5}{z^2-2z-3} dz$$

$$|z-4|=3 \Rightarrow A=3, z_0=4 \Rightarrow \alpha(4,0)$$



$$z^2-2z-3=0 \Rightarrow z_{1/2} = \frac{2 \pm \sqrt{4+12}}{2} = \frac{2 \pm 4}{2}$$

$$\Rightarrow z_1=3, z_2=5$$

$$\Rightarrow A(3,0) \in D \Rightarrow \text{pol de ord } 1.$$

$$B(-1,0) \notin D$$

$$\tilde{I} = \int_{(c)} \frac{2z+5}{z^2-2z-3} dz = \int_{c_1} \frac{\frac{2z+5}{z+1}}{z-3} dz$$

$$z-3=0 \Rightarrow z=3$$

$$a=3 \Rightarrow f(3) = \frac{1}{2\pi i} \int_{c_1} \frac{f(z)}{z-3} dz$$

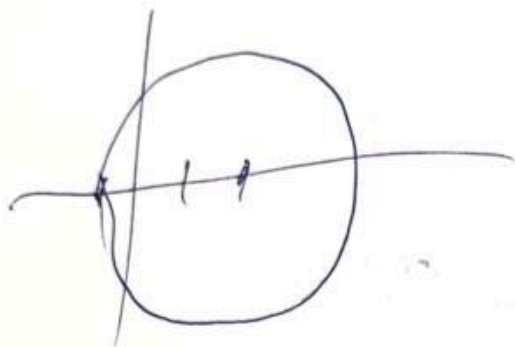
$$\frac{f(z)}{z-3} = \frac{2z+5}{z-3}$$

$$f(z) = \frac{z^2 + 5}{z - 3} \Rightarrow f(z) = \frac{z^2 + 5}{z + 1}$$

$$f(3) = \frac{6 + 5}{3 - 1} = \frac{11}{2} \Rightarrow \frac{11}{2} = \frac{1}{2\pi i} \int \frac{11 \cdot 2\pi i}{z - 3} = \frac{11 \pi i}{2}$$

$$\textcircled{b} a) \int_{|z-4|=3} \frac{z-2}{z^2+2z-3} dz$$

$$|z-2|=3 \Rightarrow z_0=2, R=3$$



$$z^2+2z-3=0 \Rightarrow z(z+1)+3(z-1)=0 \Rightarrow$$

$$\Rightarrow (z+3)(z-1)=0 \Rightarrow z_1=-3$$

$$z_2=1$$

$$\Rightarrow A(-3,0) \in D$$

$$z_2=1 \Rightarrow B(1,0) \in D \text{ pol de ord } 1.$$

$$\forall \text{ rez} \Rightarrow \Gamma = 2\pi i \text{ rez} \{f, 1\}$$

$$z=1 \text{ pol de ord } 1$$

$$\operatorname{Res}\{f, 1\} = \frac{z+1-z}{(z^2+1)(z-1)} \Big|_{z=1}$$

$$= \frac{z+1-z}{z^2+1} \Big|_{z=1} = \frac{1+1-1}{1+1} = \frac{1}{2}$$

$$\oint = 2\pi i \left(\frac{1}{2} \right) = \pi i$$